CH 7 — Directional Derivatives and Gradiant Vectors

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7.1 Directional Derivatives

How do we calcualte the rate of change of f in the directioin of some unit vector say $\hat{u}$?

If we start at (a, b) and move in the direction of $\hat{u}$, we walk along the line defined by

$$(x, y) = (a, b) + s\hat{u} \quad s \in \mathbb{R}$$

Definition — Directional Derivative

A **directional derivative** of $f(x, y)$ at a point (a, b) in the direction of the unit vector $\hat{u} = (u_1, u_2)$ is defined by

$$D_{\hat{u}} f(a, b) = \left. \frac{\mathrm{d}}{\mathrm{d}s} f(a + su_1, b + su_2) \right|_{s=0}$$

provided that the derivative exists.

Theorem — Directional Derivative and Unit Vector

$$D_{\hat{\imath}} f = f_x$$
$$D_{\hat{\jmath}} f = f_y$$

for $\hat{u} = \hat{\imath} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\hat{u} = \hat{\jmath} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

Theorem — Gradiant and Directional Derivatives

Let $g'(s) = f(a + su_1, b + su_2)$, then

$$g'(s) = \frac{\partial z}{\partial x}\frac{\mathrm{d}x}{\mathrm{d}s} + \frac{\partial z}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}s} = f_x(a + su_1, b + su_2)u_1 + f_y(a + su_1, b + su_2)u_2$$

Therefore

$$g'(0) = D_{\hat{u}} f(a, b) = f_x(a, b)u_1 + f_y(a, b)u_2 = \nabla f(a, b) \cdot \hat{u}$$

if f is differentiable and $\hat{u}$ must be a unit vector.

Example:

1. Find the directional derivative of $f(x, y) = x^2 - y^2$ at $(1, 2)$ in the direction of $\hat{u} = \frac{1}{\sqrt{2}}(1, 1)$

$$D_{\hat{u}} f(1, 2) = \left. \tfrac{\mathrm{d}}{\mathrm{d}s} f\left(1 + \tfrac{s}{\sqrt{2}}, 2 + \tfrac{s}{\sqrt{2}}\right) \right|_{s=0} = \left. \tfrac{\mathrm{d}}{\mathrm{d}s} \left[\left(1 + \tfrac{s}{\sqrt{2}}\right)^2 - \left(2 + \tfrac{s}{\sqrt{2}}\right)^2 \right] \right|_{s=0}$$
$$= \left. \tfrac{\mathrm{d}}{\mathrm{d}s} \left[1 + \sqrt{s} + \tfrac{s^2}{2} - 4 - 2\sqrt{2}s - \tfrac{s^2}{2} \right] \right|_{s=0} = \left. \tfrac{\mathrm{d}}{\mathrm{d}s} \left[-3 - \sqrt{2}s \right] \right|_{s=0} = \left. -\sqrt{2} \right|_{s=0} = -\sqrt{2}$$

2. Find the directional derivaive of $f(x,y) = x^3 + 3y^2$ at the point $(-1,2)$ in the direction of $\vec{u} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$

   $f_x = 3x^2, f_y = 6y$, continuous, thus f is differentiable

   $\hat{u} = \frac{1}{\sqrt{3^2+4^2}} = \begin{bmatrix} \frac{3}{5} \\ \frac{4}{5} \end{bmatrix}$

   $D_{\hat{u}} f(-1,2) = \nabla f(-1,2) \cdot \hat{u} = (3,12) \cdot \left(\frac{3}{5}, \frac{4}{5}\right) = \frac{57}{5}$

7.2 Greatest Rate of Change

QUESTION: What direction maximize $D_{\hat{u}} f(a,b) = \nabla f(a,b) \cdot \hat{u}$?

Recall: $\vec{u} \cdot \vec{v} = \|\vec{u}\|\|\vec{v}\| \cos\theta$ where θ is the angle between $\vec{u}$ and $\vec{v}$

Then $D_{\hat{u}} f(a,b) = \| \nabla f(a,b)\| \, \|\vec{u}\| \cos\theta = \|\nabla f(a,b)\| \cos\theta$

To maximize, we want $\theta = 0, \hat{u}$ points in the same direction $\nabla f(a,b)$, which is the direction of the steepest ascent

Theorem — Steepest Rate of Ascent

If $f(x,y)$ is differentiable at (a,b) and $\nabla f(a,b) \neq (0,0)$, then the largest value of

$$D_{\hat{u}} f(a,b) = \|\nabla f(a,b)\|$$

and it occurs when $\hat{u}$ points in the same direction as $\nabla f(a,b)$

Additionally,

$$\hat{u} = \frac{\nabla f(a,b)}{\|\nabla f(a,b)\|}$$

Question: Which direction minimizes $D_{\hat{u}} f(a,b)$?

ANS: $-\nabla f(a,b)$ is the direction of steepest descent.

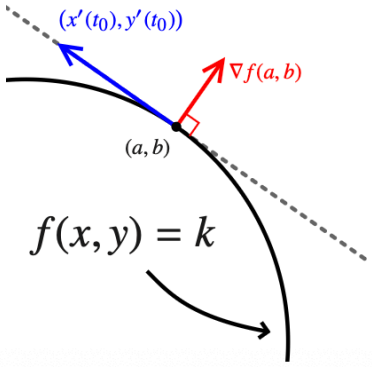
That is $\hat{u} = -\frac{\nabla f(a,b)}{\|\nabla f(a,b)\|}$

Question: Which direction should we travel if we want to stay at the same height?

ANS: $D_{\hat{u}} f(a,b) = \|\nabla f(a,b)\| \cos\theta = 0 \implies \theta = \frac{\pi}{2}$, that is orthogonal to the gradiant vector.

Theorem — Gradiant Vector and Level Curve

If $f(x,y) \in C^1$(guarantees a unique level curve through (a,b)) in some neigbourhood of (a,b) and $\nabla f(a,b) \neq (0,0)$, then $\nabla f(a,b)$ is orthogonal to the level curve which passes through (a,b)



7.3 Gradiant Vector in 3D

If f is differentiable, then we can define the directional derivative of $f(x,y,z)$ as

$$D_{\hat{u}} f(a,b,c) = \nabla f(a,b,c) \cdot \hat{u}$$

When $\hat{u}$ points in the same direction as $\nabla f(a,b,c)$, it is the steepest ascent

$D_{\hat{u}}(a,b,c) = 0$ when $\hat{u}$ is orthogonal to the gradiant vector. The gradiant vector is orthogonal to the **level surface** $f(x,y,z)$ passing through (a,b,c)

> 📘 **Definition — Level surfaces**
>
> The **level surfaces** of $f(x,y,z)$ are defined by
>
> $$f(x,y,z) = k, \quad k \in \text{ran} f$$

Note: we can defne surfaces in two ways: $z = f(x,y)$(explicitly) and $f(x,y,z) = k$(implicitly).

Notice that if $z = f(x,y)$ can be written in form $g(x,y,z) = z - f(x,y) = 0 \implies g(x,y,z) = 0$

Examples:

1. Describe the level surfaces of $f(x,y,z) = x + 2y + 3z$

   Level surface: $x + 2y + 3z = k, \quad k \in \mathbb{R}$

   A family of parallel planes with normal vector $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$

2. Describe the level surfaces of $f(x,y,z) = x^2 + y^2 + z^2$

   Level surfaces: $x^2 + y^2 + z^2 = k, \quad k \in \mathbb{R}_{\geq 0}$

   A family of a concentric spherical shells of radius $\sqrt{k}$

> 📐 **Theorem — Gradiant Vectors and Level surfaces**
>
> If $f(x,y,z) \in C^1$ in a neighborhood of $(a,b,c)$and $\nabla f(a,b,c) \cdot (0,0,0)$, then $\nabla f(a,b,c)$ is orthogonal to the level surface f$(x,y,z) = k$ through (a,b,c).

Consider a surface defined by $f(x,y,z)=k, \quad k\in\mathbb{R}$

So $\nabla f(a,b,c)$ is orthogonal to the tangent plane of $f(x,y,z)=k$ at (a,b,c) to find the equation to the tangent plane.

The vector $\vec{x}-\vec{a}=(x,y,z)-(a,b,c)$ lies on the tangent plane

$$\nabla f(a,b,c)\cdot(\vec{x}-\vec{a})=0$$

> 🐋 **Info — Equation to the Tangent plane**
>
> $$f_x(\vec{a})(x-a)+f_y(\vec{a})(y-b)+f_z(\vec{a})(z-c)$$

Examples:

1. Find the equation of the tangent plane to $z=e^xy^2+1$ at $(0,2,5)$

   $g(x,y,z)=z-e^xy^2=1$

   $\nabla g(0,2,5)=(-e^0(2)^2,-2(2)e^0,1)=(-4,-4,1)$

   $\nabla g(0,2,5)\cdot(0,2,5)=(-4,-4,1)\cdot(x-0,y-2,z-5)=-4x-4(y-2)+z-5=0$

   $\therefore z=5+4x+4(y-2)$

2. Using linearization to solve for the same equation $f(x,y)=e^xy^2+1$

   $f_x=e^xy^2, f_y=2ye^x$

   $z=f(0,2)+f_x(0,2)(x-0)+f_y(0,2)(y-2)=5+4x+4(y-2)$

   We obtain the same result as using the gradiant vector

3. Consider two surfaces, $z=0$ and $x^2+y^2+z^2=1$. Find a tangent vector to the curve of intersection at $(0,1,0)$

   $\implies x^2+y^2+0^2=1\implies x^2+y^2=1$

   Let $f(x,y,z)=x^2+y^2+z^2$ and $g(x,y,z)=z$

   Since c lies on both surfaces, $\vec{t}$ is orthogonal to both ∇f and ∇g

   $\vec{t}=\nabla f\times\nabla g=\begin{bmatrix}0\\2\\0\end{bmatrix}\times\begin{bmatrix}0\\0\\1\end{bmatrix}=\begin{bmatrix}2\\0\\0\end{bmatrix}$